

PR. 2 Logic Box

Project: Logic Box

Objective:

Develop a Python program called Pattern Generator and Number Analyzer. This project emphasizes control structures, loops (for and while), the range() function, control statements (break, continue, pass), and nested loops for generating patterns. This program allows students to practice these programming concepts and logical problem-solving skills.

Requirements:

Pattern Generator:

- Implement a feature that generates various patterns based on user input, using nested loops.
- Use for and while loops to create common pattern shapes (e.g., triangle).
- Prompt the user to enter the number of rows for each pattern.
- Demonstrate nested loops to control the rows and columns of each pattern, allowing for complex shapes.

Pattern to Implement:

- Right-angled triangle

Number Analyzer:

- Create a feature that analyzes a range of numbers, based on user input.
- Prompt the user to enter a start and end number, then use the range() function to generate numbers within this range.
- Use a for loop to iterate over this range and perform the following checks:
 - Odd or Even: Display whether each number is odd or even.
 - Sum of all Numbers: Calculate and display the sum of all numbers from the start to end range.

Control Statements:

- Use control statements effectively within loops to control the flow of the program.
- For example, use break to stop pattern generation if the user enters an invalid row

count.

- Use the pass statement as a placeholder in sections of code where logic will be added later or to skip certain numbers without an operation.

Menu-Driven Interface:

- Generate a selected pattern (right-angled triangle).
- Analyze a range of numbers with the given criteria.
- Exit the program.

Program Flow:

Welcome and Instructions:

- Display a welcome message and describe the available features of the program.

Menu Options:

- Pattern Generation: Prompt the user to select a pattern type (e.g., triangle) and enter the number of rows. Display the generated pattern using nested loops.
- Number Analysis: Prompt the user to enter a start and end range. Then display:
 - Odd and even numbers in the range.
 - Sum of all numbers in the range.
- Exit: End the program with an exit message.

Error Handling and Input Validation:

- Validate user input (e.g., ensure row count for patterns is positive and that the end of the range is greater than the start).
- Provide appropriate error messages and prompt the user to re-enter values if needed.

Exit Message:

- Thank the user for using the Student Data Organizer and displaying an exit message.

Example Console Interaction:

Welcome to the Pattern Generator and Number Analyzer!

Select an option:

1. Generate a Pattern
2. Analyze a Range of Numbers

3. Exit

Enter your choice: 1

Enter the number of rows for the pattern: 5

Pattern:

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Select an option:

1. Generate a Pattern

2. Analyze a Range of Numbers

3. Exit

Enter your choice: 2

Enter the start of the range: 10

Enter the end of the range: 15

Number 10 is Even

Number 11 is Odd

Number 12 is Even

Number 13 is Odd

Number 14 is Even

Number 15 is Odd

Sum of all numbers from 10 to 15 is: 75

Select an option:

1. Generate a Pattern

2. Analyze a Range of Numbers

3. Exit

Enter your choice: 3

Exiting the program. Goodbye!

Instructions:

Task Completion: Ensure that you attempt all the assigned tasks given to you as part of the exam. Complete each task to the best of your ability, following the instructions provided.

Assumptions: Make suitable assumptions wherever necessary, based on the requirements and instructions provided. Document any assumptions made in your project documentation or README.md file.

GitHub Repository: Create a GitHub repository to host your project. Upload your project files, including source code, and documentation to the repository. Ensure that you provide a clear and descriptive README.md file.

No Copying: Do not copy code or any other content from your classmates or any other source. Plagiarism is strictly prohibited and can result in severe consequences, including academic penalties. Ensure that all the code and content in your project are original and properly attributed to the appropriate sources, if applicable.

Submission: Once you have completed your project and uploaded it to your GitHub repository, submit the GitHub repository link to your instructor or as instructed. Double-check that your repository is properly organized and includes all the required files as per the instructions provided.

Remember to follow the instructions provided professionally, make suitable assumptions wherever necessary, and avoid copying code or content from unauthorized sources. Good luck with your project work!

Logic Box

PYTHON - DATA SCIENCE

BRING ON YOUR CODING ATTITUDE

“Quality is our Motto.”
Shaping “skills” for “scaling” higher...!!!

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